

Neural Network-Based Competency Profiling for Adaptive Engineering Education: An Ethics-by-Design Approach

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Abstract

The integration of artificial intelligence into educational assessment systems raises critical questions regarding algorithmic fairness, learner autonomy, and inclusive design. This paper presents a competency profiling system combining traditional pedagogical frameworks (T-Shaped Skills, 4C model, Bloom's Taxonomy) with predictive analytics to support adaptive educational pathways in engineering education. The system was validated through a mixed-methods study involving 1,459 students from three Russian universities, complemented by Delphi expert assessments from 20 faculty members and 20 industry employers (Kendall's $W = 0.78$; $p < 0.001$). Skill demand forecasting through 2035 was performed using an ensemble of ARIMA, Prophet, and XGBoost models (ensemble MAPE = 6.9%; $R^2 = 0.93$). Gap analysis identified critical competency deficits in programming (37.9%), process evaluation (32.2%), and solution development (30.4%). Forecasting indicates exponential growth in demand for programming and AI skills (up to 245% by 2035). Based on empirical findings, we propose a conceptual architecture for an AI-Enhanced Adaptive Tutor built on neural network models (LLM/RAG) with a Dual Knowledge Base. The study introduces an Ethics-by-Design framework ensuring transparency, contestability, fairness, and inclusivity throughout the system lifecycle.

Keywords—artificial intelligence in education, neural networks, student profiling, adaptive learning, ethical AI, algorithmic fairness, inclusive design, engineering education

I. INTRODUCTION

The rapid transition to Industry 4.0 and 5.0 has created a fundamental gap between the competencies of engineering graduates and labor market expectations. The World Economic Forum [1] reports that 44% of workers' core skills will require updating within the next five years, while automation and artificial intelligence (AI) are transforming employment structures faster than educational systems can adapt. This gap is exacerbated by a paradigmatic shift from the "human-machine" model to Human-AI collaboration (the "centaur" paradigm), where a specialist's value is increasingly determined by their ability to productively collaborate with AI [1], [2].

AI has been proposed as a means of bridging this gap through personalized educational trajectories, real-time competency assessment, and predictive analytics that anticipate future skill demand [3], [4]. However, deploying AI systems in educational contexts involves serious ethical challenges extending beyond technical metrics: algorithmic bias, constraints on learner autonomy, reproduction of structural inequalities, and data privacy concerns [5], [6].

The central idea of this work is that AI can serve not only as an object of study (AI literacy) but also as a learning tool (AI-enhanced learning). Based on data about individual students' competencies—their strengths and weaknesses—it becomes possible to construct adaptive educational pathways that replace the standardized one-size-fits-all approach. Existing uniform learning tracks in STEM education, while providing useful structural frameworks, insufficiently account for individual competency variability and learning pace [4], [7].

This study addresses three research questions:

RQ1 (Technical): What are the key competency gaps between engineering-economics students' current proficiency and labor market requirements, and how can predictive analytics inform curriculum design?

RQ2 (Ethical-Agential): What are the implications of automated competency profiling for student autonomy, and how can meaningful human agency be preserved?

RQ3 (Ethical-Inclusive): How can AI-based profiling systems be designed to support inclusive education, particularly for underrepresented students?

The contribution is threefold: (1) empirical identification of competency gaps on a large sample ($n = 1,459$) with Delphi validation and ensemble demand forecasting through 2035; (2) a conceptual architecture for an AI-Enhanced Adaptive Tutor based on LLM/RAG with a Dual Knowledge Base; (3) an Ethics-by-Design framework for responsible AI deployment in education.

II. RELATED WORK

A. AI in Education and Learning Analytics

Machine learning applications to educational data have evolved from early intelligent tutoring systems [8] to modern platforms using deep learning for student modeling and outcome prediction [2], [9]. Zawacki-Richter et al. [10] reviewed 146 empirical studies, documenting consistent benefits for personalized feedback. Khosravi et al. [3] demonstrated that AI-based adaptive systems enhance engagement in STEM disciplines. However, Holmes et al. [2] note that most research originates from well-resourced institutions in high-income countries, raising generalizability concerns. Chen et al. [7] further identified publication bias favoring positive outcomes.

B. Competency-Based Education and the Labor Market

Competency-based approaches have become the dominant paradigm for aligning higher education with labor market needs [11]. Models such as T-Shaped Skills, 4C (critical thinking, creativity, communication, cooperation), and Bloom's Taxonomy provide structured assessment frameworks [12], [13]. However, employer-defined competencies often reflect short-term organizational needs rather than long-term graduate adaptability [14]. Timiras et al. [15] found that technical competencies dominate hiring criteria, while soft skills receive substantially less weight in actual hiring decisions.

C. Ethical AI and Inclusive Design in Education

Research on algorithmic fairness and accountability is expanding [16], [17], yet education receives comparatively less attention than criminal justice or credit scoring [18]. O'Neil [6] demonstrates how algorithmic systems create feedback loops amplifying historical inequalities. In education, these risks manifest through recommendation algorithms reproducing professional segregation [19]. Universal Design for Learning (UDL) principles [20], [21] provide a foundation for inclusive design, while Jang et al. [22] showed the need for person-oriented approaches accounting for intra-individual variability. This work positions ethical considerations as constitutive of design, following value-sensitive design methodologies [23].

III. METHODOLOGY

A. Study Design and Sample

The study employed a convergent mixed-methods design combining quantitative surveying, Delphi expert assessment, and predictive modeling. The sample comprised 1,459 students in engineering-economics programs from three Russian universities: Peter the Great St. Petersburg Polytechnic University (SPbPU, $n = 612$; 41.9%), Russian Presidential Academy of National Economy and Public Administration (RANEPA, $n = 485$; 33.2%), and Ural State University of Economics (USUE, $n = 362$; 24.8%). The age distribution was 18–24 years (55.6%), 25–30 (28.3%), 31–40 (11.8%), and over 40 (4.3%). The gender ratio was 1:2.8 (male:female). A majority of respondents (62.4%) combined studies with employment. All participants provided informed consent; data were processed anonymously.

B. Competency Assessment Instrument

The assessment instrument comprised 28 items organized across three validated competency models: T-Shaped Skills (engineering depth, digital literacy, project management, programming), 4C model (critical thinking, creativity, communication, cooperation), and Bloom's Taxonomy (from concept understanding to process evaluation and improvement). Each item was rated on a 5-point Likert scale (1—very low to 5—very high). The instrument demonstrated acceptable internal consistency (Cronbach's $\alpha = 0.87$).

C. Delphi Validation and Gap Analysis

The Delphi method was implemented with two expert panels: 20 academics (faculty from the three participating universities) and 20 employers (from technology, industrial, and financial-economic sectors). Three iterative rounds achieved convergence, with a final Kendall's $W = 0.78$ ($\chi^2 = 405.6$; $p < 0.001$), indicating statistically significant high

agreement. Gap Scores were calculated as: $GS = (\text{Required Level} - \text{Current Level}) / \text{Required Level} \times 100\%$, where the required level was based on employer consensus (Delphi round 3) and the current level was the equally weighted mean of three respondent groups. Gaps exceeding 30% were deemed critical.

D. Forecasting Models

An ensemble of ARIMA, Prophet [25], and XGBoost models was employed to forecast competency demand through 2035, based on HeadHunter job posting data for 2020–2025. XGBoost incorporated additional features including industry trends, technological cycles, and macroeconomic indicators. The forecast horizon was divided into short-term (2025–2027), medium-term (2028–2030), and long-term (2031–2035) periods.

IV. RESULTS

A. Competency Profile and Gap Analysis

Self-assessment analysis ($n = 1,459$) across 14 competencies revealed pronounced unevenness. Highest mean scores were obtained for cooperation (4.35) and concept understanding (4.18), indicating effective development of soft skills in the university environment. Lowest scores were recorded for programming and automation (2.94) and process evaluation (2.98), revealing critical gaps in higher-order technical competencies.

Delphi validation revealed a consistent pattern: faculty rated student competencies higher than students themselves (by 0.13–0.34 points), while employers provided the most conservative assessments. The largest faculty–employer discrepancy was observed in programming (3.18 vs. 2.65; $\delta = 0.53$) and process evaluation (3.32 vs. 2.85; $\delta = 0.47$), indicating a misalignment between academic assessment standards and market requirements.

TABLE I. GAP ANALYSIS RESULTS

Competency	Required	Current	Gap (%)
Programming & Automation	4.70	2.92	37.9
Process Evaluation	4.50	3.05	32.2
Solutions Development	4.60	3.20	30.4
Project Management	4.35	3.12	28.3
Knowledge Application	4.55	3.47	23.7
Digital Literacy	4.65	3.79	18.5
Cooperation	4.45	4.34	2.5

Bold = critical gap (>30%). Required level: employer consensus (Delphi round 3). Current level: equally weighted mean of students, faculty, and employers.

Table I and Fig. 1 present the results. Three competencies exceeded the 30% critical threshold: programming and automation (37.9%), process evaluation (32.2%), and solutions development (30.4%). Project management (28.3%) approached criticality. The minimal gap was recorded for cooperation (2.5%), confirming effective development of this competency within the university environment. Cross-institutional comparison revealed variability within 0.15–0.26 points.

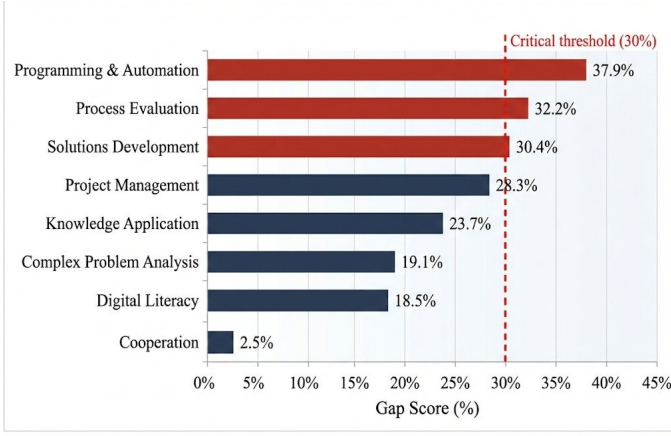


Fig. 1. Competency gap scores between current student proficiency and employer-defined requirements. Dashed line indicates the 30% critical threshold.

B. Competency Demand Forecasting

The weighted ensemble achieved the best predictive performance (MAPE = 6.9%; $R^2 = 0.93$), outperforming individual ARIMA (MAPE = 9.2%; $R^2 = 0.86$), Prophet (MAPE = 8.7%; $R^2 = 0.88$), and XGBoost (MAPE = 7.8%; $R^2 = 0.91$) models. As shown in Fig. 2, forecasts indicate exponential growth in programming and AI demand (245% by 2031–2035 relative to the 2024 baseline), followed by data analysis (198%) and interdisciplinary competencies (195%). Project management is projected at 167%, digital literacy at 155%. Moderate growth for critical thinking (138%) confirms its status as a foundational but non-differentiating competency. It should be emphasized that a 10-year forecast based on a 6-year observation window is scenario-based and requires cautious interpretation.

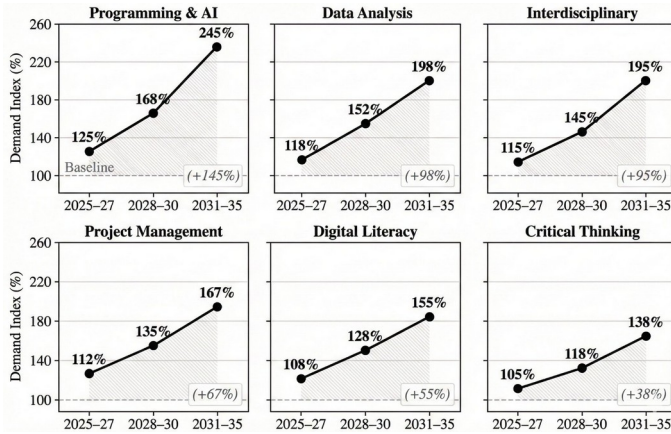


Fig. 2. Ensemble-forecasted competency demand index for six key competencies through 2035 (baseline 2024 = 100%). Net gains shown in parentheses. Ensemble MAPE = 6.9%, $R^2 = 0.93$.

V. DISCUSSION

A. Conceptual Architecture of the AI-Enhanced Adaptive Tutor

The identified competency gaps and demand dynamics substantiate the need for a systemic transition from uniform to adaptive education. We propose a conceptual architecture for an AI-Enhanced Adaptive Tutor based on Large Language

Models (LLMs), capable of personalizing educational pathways based on individual competency profiles (Fig. 3).

The architecture's core is a Dual Knowledge Base: (a) a Domain Knowledge Base—a verified disciplinary corpus implemented via Retrieval-Augmented Generation (RAG), minimizing model hallucinations and ensuring factual accuracy; and (b) a Learner Profile—a dynamic student model incorporating current competency gaps, learning history, error patterns, learning pace, and explanation preferences.

The system calibrates explanations to the student's current level: for a student with a critical programming gap (37.9%), the tutor may offer step-by-step introductions with concrete examples and visualizations, while for a student with high cooperation proficiency (2.5% gap), it may suggest integrative project-based tasks. This approach potentially addresses the fundamental problem of mass education—scaling individualized support. The architecture is presented as a conceptual design contribution; technical validation remains a subject for future work.

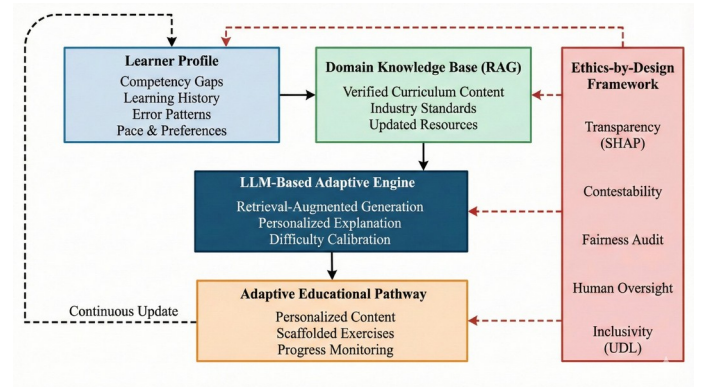


Fig. 3. Conceptual architecture of the AI-Enhanced Adaptive Tutor with Dual Knowledge Base and Ethics-by-Design layer.

B. Ethics-by-Design Framework

Since the system profiles competencies and influences educational trajectories, ethical considerations must be embedded throughout the design lifecycle [23], rather than added post-deployment. We propose a five-component framework:

Transparency: Students must understand why a particular trajectory is recommended. Explainable AI methods such as SHAP [24] enable visualization of the factors driving recommendations. **Contestability:** Students retain the right to challenge and override algorithmic recommendations, preserving human agency in educational decision-making [5], [6]. **Fairness:** Regular auditing of algorithmic outputs for systematic bias across gender, socioeconomic, and other dimensions [16], [17]. **Human Oversight:** Continuous faculty involvement in reviewing and validating system recommendations. **Inclusivity:** Design supporting diverse learners—with varying pace, educational backgrounds, and physical abilities—following UDL principles [20], [21]. Additional mechanisms include data minimization and transparent governance of student profile data.

VI. CONCLUSION

This study ($n = 1,459$; 3 universities; Delphi validation with 40 experts, $W = 0.78$) identified critical competency gaps in

engineering education: programming (37.9%), process evaluation (32.2%), and solutions development (30.4%). Ensemble forecasting (MAPE = 6.9%, $R^2 = 0.93$) indicates exponential growth in programming and AI demand to 245% by 2035.

Based on empirical results, we proposed: (1) a conceptual AI-Enhanced Adaptive Tutor architecture based on LLM/RAG with a Dual Knowledge Base enabling personalized educational trajectories; and (2) an Ethics-by-Design framework (transparency, contestability, fairness, human oversight, inclusivity) for responsible AI deployment in education.

Limitations include the self-assessment nature of competency data requiring additional objective validation; the scenario-based character of long-term forecasts given a 6-year observation window; and the need for prototyping and empirical testing of the proposed tutor architecture. Future directions include longitudinal validation through graduate career tracking, pilot prototyping of the AI tutor, and cross-institutional replication studies.

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